

An Introduction to Hypothesis Testing

Lecture Notes: Statistics for Data Science

Zamir Hussain
Department of Mathematics
University of Wah

January 9, 2026

Contents

Part I: Foundations of Hypothesis Testing	3
1 Introduction: Why is Hypothesis Testing Important?	3
1.1 The Core Process (Inferential Statistics)	3
1.2 Common Statistical Tests	3
2 Basic Concepts in Hypothesis Testing	3
2.1 Key Terminology	3
2.2 Simple Example: Testing a Fair Coin	4
Part II: Z-Test Methodology	5
3 Z-Test: When Population Standard Deviation is Known	5
3.1 Decision Criteria: Z-Test vs T-Test	5
3.2 Z-Test Formula	5
3.3 Example: Height Comparison	5
3.4 Common Z-scores for Confidence Levels	6
Part III: T-Test Methodology	7
4 T-Test: When Population Standard Deviation is Unknown	7
4.1 The T-Distribution	7
4.2 Degrees of Freedom	7
4.3 T-Test Formula	7
4.4 Example: Medication Effect on IQ	7
Part IV: Chi-Square Test for Categorical Data	10
5 Chi-Square Goodness of Fit Test	10
5.1 Conceptual Example: Bike Color Preference	10
5.2 Chi-Square Test Formula	10
5.3 Detailed Example: Weight Distribution Over Time	10
Practice Problems	12
5.4 Z-Test Problems	12
5.5 T-Test Problems	12
5.6 Chi-Square Problems	12
Appendix	13
5.7 Common Critical Values	13
5.8 Formulas Summary	13
5.9 7-Step Hypothesis Testing Procedure	13

Part I: Foundations of Hypothesis Testing

1 Introduction: Why is Hypothesis Testing Important?

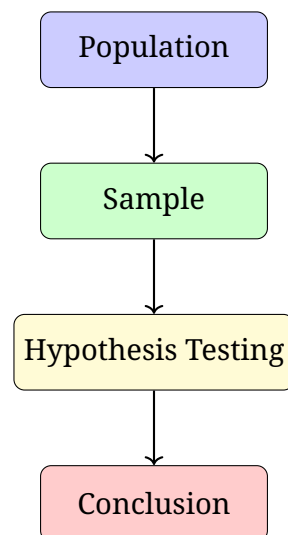
Hypothesis testing is a foundational concept in statistics and a critical skill for data analysts and data scientists. It provides a formal framework for making decisions and drawing conclusions about a larger population based on limited sample data.

Key Concept

We use data from a small **sample** to test an assumption (a hypothesis) about an entire **population**.

1.1 The Core Process (Inferential Statistics)

The process follows a logical flow:



1.2 Common Statistical Tests

- **Z-Test:** When population standard deviation is known
- **T-Test:** When population standard deviation is unknown
- **ANOVA:** For comparing multiple group means
- **Chi-Square Test (χ^2):** For categorical data analysis

2 Basic Concepts in Hypothesis Testing

2.1 Key Terminology

- **Population Parameter:** Characteristic of entire population (e.g., μ)
- **Sample Statistic:** Characteristic calculated from sample (e.g., $\bar{x}$)
- **Null Hypothesis (H_0):** Default assumption being tested
- **Alternative Hypothesis (H_1):** What we suspect might be true

- **Significance Level (α):** Probability of Type I error
- **Confidence Interval:** Range containing population parameter with certain confidence

2.2 Simple Example: Testing a Fair Coin

Example

Problem: Test if a coin is fair (equal probability of heads and tails).

Steps:

1. Hypotheses:

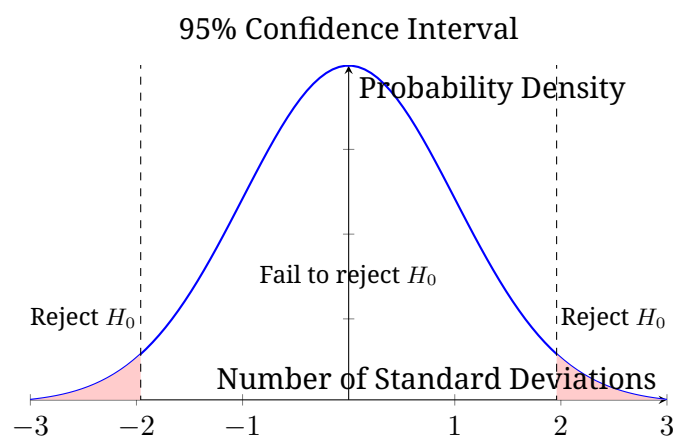
- $H_0 : P(\text{Heads}) = 0.5$ (coin is fair)
- $H_1 : P(\text{Heads}) \neq 0.5$ (coin is biased)

2. Experiment: Toss coin 100 times

3. Decision Boundary: Based on 95% confidence interval

- If heads between 40-60: Fail to reject H_0
- If heads <40 or >60 : Reject H_0

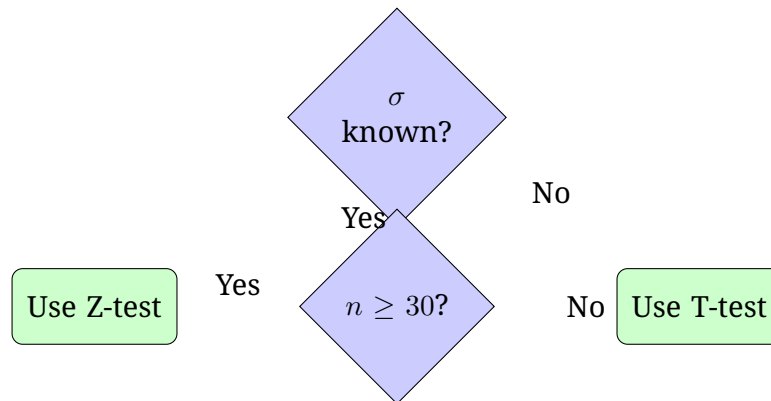
4. Significance Level: $\alpha = 0.05$



Part II: Z-Test Methodology

3 Z-Test: When Population Standard Deviation is Known

3.1 Decision Criteria: Z-Test vs T-Test



3.2 Z-Test Formula

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Where:

- $\bar{x}$ = sample mean
- μ = population mean under H_0
- σ = population standard deviation
- n = sample size

3.3 Example: Height Comparison

Example

Problem: City residents' average height is 168 cm ($\sigma = 3.9$ cm). A doctor believes it's different. Sample of 36 people has mean 169.5 cm. Test at 95% confidence.

Mathematical Solution

Solution:

Step 1: Given Information

- $\mu = 168$ cm, $\sigma = 3.9$ cm
- $\bar{x} = 169.5$ cm, $n = 36$
- $\alpha = 0.05$ (95% confidence)

Step 2: Hypotheses

- $H_0 : \mu = 168$ cm
- $H_1 : \mu \neq 168$ cm (two-tailed test)

Step 3: Calculate Z-statistic

$$Z = \frac{169.5 - 168}{3.9/\sqrt{36}} = \frac{1.5}{0.65} = 2.3077 \approx 2.31$$

Step 4: Critical Value For $\alpha = 0.05$ (two-tailed), $Z_{critical} = \pm 1.96$

Step 5: Decision Since $2.31 > 1.96$, reject H_0

Conclusion: Evidence supports doctor's claim that mean height is different.

3.4 Common Z-scores for Confidence Levels

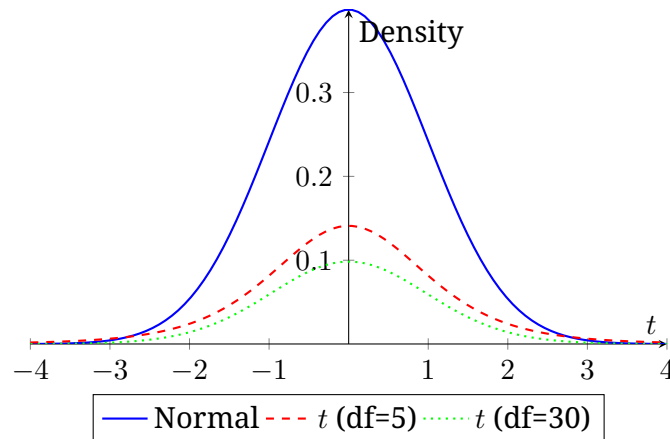
Confidence Level	α	$\alpha/2$	Two-tailed Z	One-tailed Z
90%	0.10	0.05	± 1.645	± 1.282
95%	0.05	0.025	± 1.96	± 1.645
99%	0.01	0.005	± 2.576	± 2.326

Part III: T-Test Methodology

4 T-Test: When Population Standard Deviation is Unknown

4.1 The T-Distribution

Comparison: Normal vs T-distributions



4.2 Degrees of Freedom

$$df = n - 1$$

Intuition: If you have 3 chairs and 3 people:

- First person: 3 choices
- Second person: 2 choices
- Third person: 1 choice
- Independent choices = $2 = 3-1$

4.3 T-Test Formula

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where s = sample standard deviation.

4.4 Example: Medication Effect on IQ

Example

Problem: Population IQ = 100. New medication given to 30 participants. Sample mean = 140, sample standard deviation = 20. Test if medication affects IQ at 95% confidence.

Mathematical Solution

Solution:

Step 1: Given Information

- $\mu = 100, \bar{x} = 140, s = 20, n = 30$
- $\alpha = 0.05, \text{df} = 29$

Step 2: Hypotheses

- $H_0 : \mu = 100$ (no effect)
- $H_1 : \mu \neq 100$ (two-tailed test)

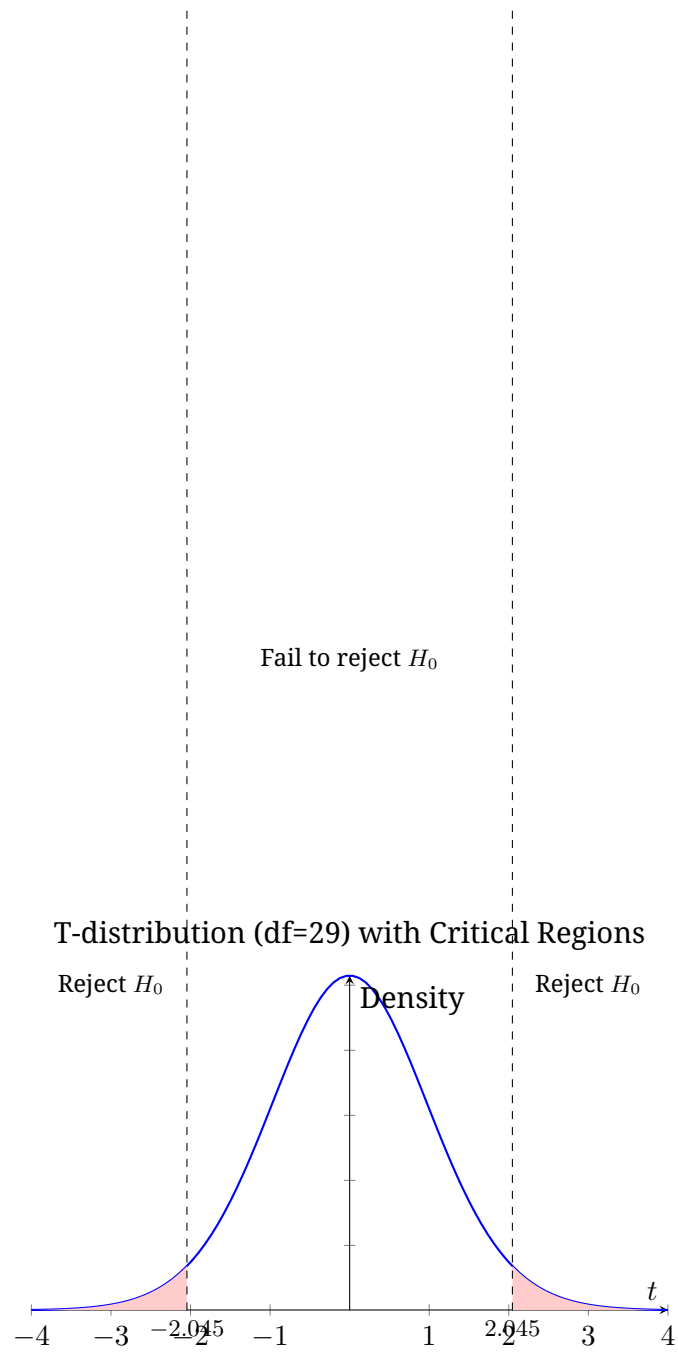
Step 3: Calculate t-statistic

$$t = \frac{140 - 100}{20/\sqrt{30}} = \frac{40}{3.65} = 10.96$$

Step 4: Critical Value For $\text{df} = 29, \alpha = 0.05, t_{\text{critical}} = \pm 2.045$

Step 5: Decision Since $10.96 > 2.045$, reject H_0

Conclusion: Medication significantly affects IQ.



Part IV: Chi-Square Test for Categorical Data

5 Chi-Square Goodness of Fit Test

5.1 Conceptual Example: Bike Color Preference

Example

Scenario: Test if bike color preferences are equal among Yellow, Orange, and Red.

- **Expected:** Equal preference (1/3 each)
- **Observed (sample of 98):** Yellow: 22, Orange: 17, Red: 59

Chi-Square test checks if the difference between observed and expected is significant.

5.2 Chi-Square Test Formula

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where:

- O = Observed frequency
- E = Expected frequency
- $df = k - 1$ (k = number of categories)

5.3 Detailed Example: Weight Distribution Over Time

Example

Problem: 2010 census showed weight distribution as: <50kg: 20%, 50-75kg: 30%, >75kg: 50%. 2020 sample of 500 people shows: <50kg: 140, 50-75kg: 160, >75kg: 200. Test if distribution changed at $\alpha = 0.05$.

Mathematical Solution

Solution:

Step 1: Calculate Expected Frequencies

$$\begin{aligned}E_{<50} &= 500 \times 0.20 = 100 \\E_{50-75} &= 500 \times 0.30 = 150 \\E_{>75} &= 500 \times 0.50 = 250\end{aligned}$$

Step 2: Calculate χ^2

$$\begin{aligned}\chi^2 &= \frac{(140 - 100)^2}{100} + \frac{(160 - 150)^2}{150} + \frac{(200 - 250)^2}{250} \\&= \frac{40^2}{100} + \frac{10^2}{150} + \frac{(-50)^2}{250} \\&= 16 + 0.667 + 10 = 26.667\end{aligned}$$

Step 3: Degrees of Freedom

$$df = 3 - 1 = 2$$

Step 4: Critical Value For $df = 2$, $\alpha = 0.05$: $\chi^2_{critical} = 5.991$

Step 5: Decision Since $26.667 > 5.991$, reject H_0

Conclusion: Weight distribution has significantly changed since 2010.

Practice Problems

5.4 Z-Test Problems

1. **Battery Life:** Company claims batteries last 100 hours ($\sigma = 10$ hours). Sample of 50 batteries averages 97 hours. Test claim at 5% significance.
2. **Bulb Warranty:** Factory claims bulbs last 5 years ($\sigma = 0.5$ years). Worker tests 40 bulbs finding average 4.8 years. Test at 2% significance if bulbs last less than claimed.
3. **Cereal Boxes:** Company claims boxes contain 500g ($\sigma = 20$ g). Consumer group samples 36 boxes finding average 490g. Test at 5% significance if boxes are under-filled.

5.5 T-Test Problems

1. **Teaching Method:** New teaching method tested on 20 students (mean score = 85, standard deviation = 10). Traditional method population mean = 80. Test if new method is better at:
 - a) 5% significance
 - b) 1% significance
 - c) Calculate effect size
2. **Light Bulbs:** Company claims bulbs last 1000 hours. Consumer group tests 15 bulbs (mean = 950 hours, standard deviation = 100 hours). Test if bulbs last less than claimed at 5% significance.
3. **Medication Assignment:** Using original medication example (population IQ = 100, sample: $n=30$, $\bar{x} = 140$, $s=20$), test with 80% confidence interval ($\alpha = 0.20$).

5.6 Chi-Square Problems

1. **Biased Die:** Test if a 6-sided die is fair. Rolled 60 times with results: 1:5, 2:8, 3:9, 4:8, 5:10, 6:20. Use $\alpha = 0.05$.
2. **Ice Cream Flavors:** Shop claims equal sales among 4 flavors. Sample of 200 sales: Vanilla:60, Chocolate:55, Strawberry:45, Mint:40. Test claim at 95% confidence.
3. **Political Poll:** Historical data: 40% Democrat, 40% Republican, 20% Independent. New poll of 400 voters: Democrat:140, Republican:180, Independent:80. Test for significant shift at $\alpha = 0.05$.

Appendix

5.7 Common Critical Values

df	80% CI $\alpha = 0.20$	90% CI $\alpha = 0.10$	95% CI $\alpha = 0.05$	98% CI $\alpha = 0.02$	99% CI $\alpha = 0.01$
1	3.078	6.314	12.706	31.821	63.657
5	1.476	2.015	2.571	3.365	4.032
10	1.372	1.812	2.228	2.764	3.169
20	1.325	1.725	2.086	2.528	2.845
29	1.311	1.699	2.045	2.462	2.756
30	1.310	1.697	2.042	2.457	2.750
∞	1.282	1.645	1.960	2.326	2.576

Table 1: Two-tailed critical t-values

5.8 Formulas Summary

Mathematical Solution

$$\text{Z-test: } Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

$$\text{T-test: } t = \frac{\bar{x} - \mu}{s/\sqrt{n}}, \quad df = n - 1$$

$$\text{Chi-Square: } \chi^2 = \sum \frac{(O - E)^2}{E}, \quad df = k - 1$$

$$\text{Confidence Interval (Z): } \bar{x} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

$$\text{Confidence Interval (t): } \bar{x} \pm t_{\alpha/2} \times \frac{s}{\sqrt{n}}$$

$$\text{Type I Error: } \alpha = P(\text{Reject } H_0 | H_0 \text{ true})$$

$$\text{Type II Error: } \beta = P(\text{Fail to reject } H_0 | H_0 \text{ false})$$

5.9 7-Step Hypothesis Testing Procedure

1. **State hypotheses:** H_0 and H_1
2. **Choose significance level:** α (typically 0.05)
3. **Select appropriate test:** Z-test, T-test, or Chi-square
4. **Calculate test statistic**
5. **Determine critical value**
6. **Make decision:** Reject or fail to reject H_0
7. **Draw conclusion in context**